Kaggle Competition: Facial Feature Classification

ENGE707 Data Engineering and AI

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1. Dataset Exploration and Understanding

1.1 Problem



The problem focuses on distinguishing between cats and dogs and which animal facial features (eye, nose, mouth or face) an image shows. This is a classification problem where there are 8 different classes: CatEyes, DogEyes, CatFace, DogFace, CatMouth, DogMouth, CatNose, or DogNose that the image can be classified as. If the model can succeed in providing a high accuracy and other metrics in the prediction results, the model could be used as the first step to further classify the breed and visual anomalies of the cat / dog. This prediction would allow for example a system to automatically describe what section of the animal is being checked and check the region for possible anomalies.

1.2 Data Inspection

```
Train rows, columns: (63660, 2)
Column dtypes: {'image_id': 'str', 'class': 'str'}
Null values: {'image_id': 0, 'class': 0}
Duplicate ids in CSV: 0
Train id range: 1 - 63660
Train file types: {'.jpg': 63660}

Test rows, columns: (29443, 2)
Duplicate ids in sample_submission: 0
Test id range: 1 - 29443
Test file types: {'.jpg': 29443}
Test classes in sample_submission: {'CatEye': 4567, 'DogEye': 4221, 'CatFace': 3571, 'CatMouth': 3570, 'CatNose': 3457, 'DogFace': 3407, 'DogMouth': 3341, 'DogNose': 3309}

Train files: 63660 | Test files: 29443
```

Figure 1.2.1 : Dataset Structure

The Kaggle competition provided testing and training folders as well as included the train_labels.csv which contains the mapping of the “image_id” and “class” columns, the “image_id” 00001 links to 00001.jpg in the train folder and the class is the target. The test folder contains the unlabelled testing images and the train folder contains the labelled training images (connected to the train_labels.csv).

In the training dataset, there are 63,660 rows and in the testing dataset there are 29,443 rows. The train_labels.csv consists of 2 columns “image_id” and “class”. With “image_id” as the unique identification of the image and class being the target.
In Figure 1.21, it showed that there are 0 duplicate ids in CSV and the id correctly ranges from 1 to 63660 which is the correct amount of images in the training folder.

The sample_submission.csv shows what the structure of the data submission to kaggle would need to be like with an unpadded “image_id” (e.g 00001 to 1) and the class column that describes the predicted target attribute.

1.3 Number of Instances in each class

	class	instances
0	CatEye	13884
1	CatFace	6942
2	CatMouth	6942
3	CatNose	6942
4	DogEye	11580
5	DogFace	5790
6	DogMouth	5790
7	DogNose	5790

Figure 1.3.1 : Class Distribution

In the train_labels.csv there, it shows that not all classes are distributed evenly. With CatEye (13,884) being the dominant class having more than twice the amount of instances than the other classes such as DogFace, DogMouth, and DogNose (5,790). This shows that there is a class imbalance in the training dataset. This would be considered during the preprocessed stage where the training and validation split.

1.4 Class Image Inspection and Evaluation



Figure 1.4.1 : Example Classification Images

The images shown of each class (CatEye, CatFace, CatMouth, CatNose, DogEye, DogFace, DogMouth, and DogNose) all showed varying infomation. The images shown appeared to vary in resolution with some seemingly more pixelated than others, Some images do not clearly show the features such as the DogFace id17364 where it is hard to view the dog's face due to the fur color and clarity.

Additionally some of the images shown contain other classes that they weren't supposed to capture such as the DogNose id 27348 showed the Nose as well and 1 eye. This would make classification for models much more difficult due to the combination of multiple target classes.

The lighting and position of the images are also different as some appear overly bright and some appear dark. The head position of the cats and dog images are also not perfectly horizontal (e.g the CatNose Id 62280).

Image dimension summary:

	width			height		
	min	median	max	min	median	max
split						
test	8	82.0	1832	9	80.0	1978
train	4	84.0	2219	4	80.0	2219

Figure 1.4.2 : Image Dimension Summary

Further confirming the dimension of the images, Figure 1.4.2 shows the summary of all the images, with the train images ranging from 4px to 2219px and the height ranging from 4px to 2219px. This range is apparent for the test images as well.

```
File-reading or decoding failures: 0
```

Figure 1.4.3 : Checking Corrupted Files

Confirmed that there are no corrupted files in both the testing image folder and the training image folder.

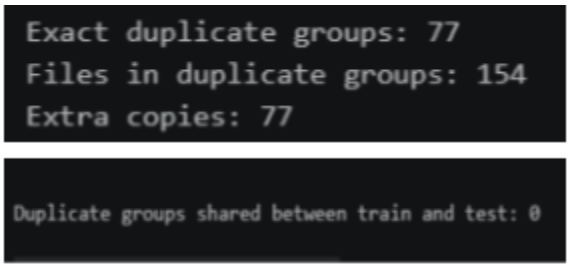


Figure 1.4.4 : Checking Duplicates

There are 77 exact duplicate groups from using SHA 256 hash in both the testing folder and the training folder. This means that there are 77 different groups of images, and 2 images have the same hash. Though there are duplicates in both the testing and training folder, there are no shared duplicates between those folders. Therefore to prevent data leakage, the duplicates would need to be kept in the same dataset for training or validation. This checks only if the byte content is the same, it does not check if the image was resized, or recompressed and each image has not been checked visually if they are similar. Further evaluation would proceed in the preprocessing stage whether duplicates would need to be removed.

1.5 Explanation of usage of Training, Validation and Kaggle Submission

For the dataset, the test dataset (images) would not be used in training in any way, only used for the testing for the final kaggle submission. The training dataset (images) would be applied to the 80 / 20 split so that 80% of the data would be for training and the 20% would be for validation. Stratified splitting would be applied to ensure that the class proportion is the same for both the training and the validation subsets. The 80% training data would be used to actually learn the pattern from images to adjust appropriate weights for each classification and the validation dataset is used to evaluate the performance on unseen data, so that further improvement could be made. Once the performance is satisfactory the trained deep learning model would be used on the test dataset to predict all instances in the dataset, the result of the prediction would be saved in a csv file to be submitted to kaggle.

2. Data Preprocessing and Pipeline Preparation

In the preprocessing stage I have decided to use a pretrained CNN (Convolutional Neural Networks) model called ResNet18. This preprocessing pipeline would be based on trying to optimize the images so it would perform the best on the defined model. Further explanation on why I chose this model and architecture would be in section 3 (Mode selection and pipeline development).

2.1 Data Preparation

To prepare the data for the other preprocessing stage (training-validation split, image resize, etc.). The training dataset ensured that the train_label.csv file had an appropriate mapping between each image_id and its image path.

```
Training duplicate groups: 15
Files in duplicate groups: 30
Extra duplicate copies: 15
```

Figure 2.1.1 : Duplicates in training dataset

```
Duplicate groups with conflicting labels: 0
Empty DataFrame
Columns: [image_id, class, path, sha256]
Index: []
```

Figure 2.1.2 : Checking Duplicates conflicting labels

In Task 1, it was identified that there are 77 exact duplicate groups using hash; therefore, another check has been performed in the training dataset to see the number of duplicate groups. The SHA-256 identified that there are 15 duplicate groups, and no conflicting labels were found; therefore, the images are duplicates. The duplicate images are removed from the training dataset because splitting the dataset into two 2 subsets, training and validation, could potentially have the same image appear on both the training and validation sets. This would cause data leakage, where they memorise the images and provide a falsely high validation accuracy. The duplicates were removed from the training dataframe, and the total training dataset went from 63,660 to 63,645 images.

```
Before: 63660
After: 63645
Removed: 15
```

Figure 2.1.3 : Summary of training dataset after removal

2.2 Training and Validation Split

```
Training images: 50916
Validation images: 12729
Total: 63645
```

class	Original (%)	Training (%)	Validation (%)
CatEye	21.80	21.80	21.80
CatFace	10.91	10.91	10.91
CatMouth	10.91	10.91	10.90
CatNose	10.91	10.91	10.90
DogEye	18.19	18.19	18.19
DogFace	9.09	9.09	9.10
DogMouth	9.10	9.10	9.10
DogNose	9.10	9.10	9.10

```
No overlapping image IDs or exact duplicate hashes.
```

Figure 2.2.1 : 80/20 split Confirmation

The 80/20 split method was used to split 80% to the training subset (50,916 images) and 20% to the validation subset (12,729 images). This allows the model to train on the majority of the total dataset and have enough unseen data to validate. To maintain the same amount of proportions across both subdatasets, stratification is used to ensure that the class proportions are approximately similar relative to the proportions of the original dataset. This is to prevent overrepresentation of a class in one dataset rather than the other from the original dataset. A seed of 42 is used to ensure this split is reproducible.

2.3 Resize Image



Figure 2.3.1 : 6 Examples on resizing applied

From the investigation of the dataset, we can see that the images vary in size, whereas the pretrained CNN model uses a standard image size (224 x 224). Therefore, images are resized proportionally to 224 x 224 if possible, and padding is applied to fill up the image to the set standard image size; this is to ensure that the original aspect ratio is preserved using padding, unlike resize methods such as cropping, which may remove important facial features. For larger images, it is scaled down proportionally, and padding is added to reach the set size stated.

For smaller images, upscaling the image generally preserves details; scaling down for larger images may remove finer image details but is necessary to match the set image dimensions. The main benefit for this type of resizing is that the image framing is maintained and does not introduce distortion of the image.

In Figure 2.3.1, the first image padding is applied to the image because the image can't scale properly to fit 224x224; therefore, black bars are added to the top and bottom, and the image does not show any distortion.

2.4 Scale and Normalisation of Pixel

```
Image class: CatNose
Original image size: (224, 224)

After rescaling:
Tensor shape: torch.Size([3, 224, 224])
Minimum value: 0.0
Maximum value: 1.0

After normalisation:
Tensor shape: torch.Size([3, 224, 224])
Minimum value: -2.1179039478302
Maximum value: 2.5354249477386475
```

Figure 2.4.1 : A example of scaling and normalisation

The image pixel values first consist of values from 0- 255, as they follow the RGB scale. It is rescaled to 0 - 1 to ensure that the image pixel values are compatible with the CNN model, as the input values are on a more consistent numerical scale than the original pixel values.

After scaling the pixel values are then normalised because the ResNet18 model follows a set standard deviation and mean. The standard deviation for red is 0.229, green 0.224, and 0.225 for blue. The mean is 0.485 for red, 0.456 for green, and 0.406 for blue. These values are used to normalize since the pretrained ResNet18 model was trained using these preprocessing values. Figure 2.4.1 shows an example of an image being scaled to 0-1 and normalised to a value range of -2.11 to 2.53.

2.5 Encoding

```
Class mapping: {'CatEye': 0, 'CatFace': 1, 'CatMouth': 2, 'CatNose': 3, 'DogEye': 4, 'DogFace': 5, 'DogMouth': 6, 'DogNose': 7}

Training examples:
  image_id     class  label
0    26977   CatNose      3
1    43790  DogMouth      6
2    60045    CatEye      0
3    47430    DogEye      4
4    50148  DogMouth      6
```

Figure 2.5.1 : Integer Encoding Evidence

Encoding is used to turn text labels into numbers in order for the model to map the classes easily. Integer encoding is utilized since the 8 different classes can be represented with unique values, and these values do not need to be represented in order. This method selects a unique identifier value to assign to a class; in Figure 2.5.1, CatEye is 0, CatFace is 1, CatMouth is 2, CatNose is 3, DogEye is 4, DogFace is 5, DogMouth is 6, and DogNose is 7.

(Note: For submission, the predicted labels will be converted back to their corresponding original labels for Kaggle submission.)

2.6 Data Augmentation

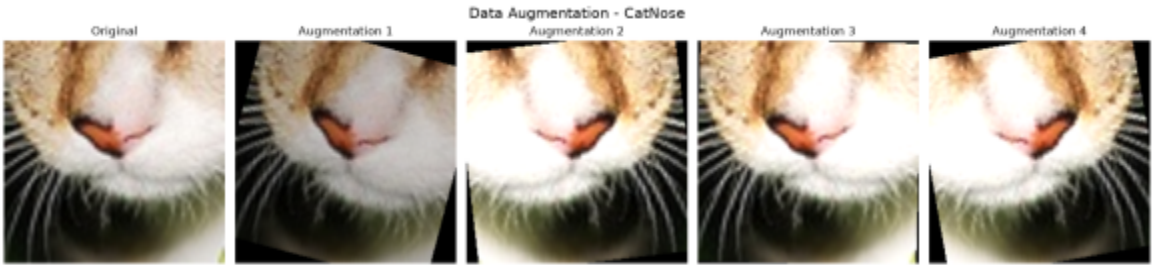


Figure 2.6.1: Example Augmentation

In the task, during the data exploration, it was concluded that there is much variation in the images, such as brightness, orientation of animal features, etc. This means that the model would recognize facial features under a wide variation of image conditions. Augmentation allows the model to improve generalization for unseen images and reduce overfitting. The data had been augmented by introducing horizontal flipping, rotations, and brightness adjustments; these augmentations are randomly applied to the training subset, and different variations of augmentation could be generated during training. In Figure 2.6.1, the first image shows the original, while the other images show the applied augmentations, with Augmentation 1 rotated to the right and lowered in brightness. Augmentation 4 shows the image rotated left, with increased brightness and horizontally flipped. This is not applied to the testing and validation datasets, as these datasets need to be kept unchanged from augmentation and only undergo other preprocessing stages, as the images need to be consistent to evaluate results.

2.7 Data Loaders

Data loaders are used in the preprocessing as this is to ensure that not all images are loaded at one time. This makes sure that the entire dataset isn't stored in the GPU memory. The batch size used is initially 32 at this stage as this is a good recommended starting point between efficiency and memory consumption (This could be changed later on depending on the memory usage, efficiency and validation results) . The images are shuffled in the training subset to allow the model to encounter different orders of images, for example the image could encounter thousands of the same class before others thus the weights would temporarily be more biased towards the most recently dominated class. While the training subset is shuffled the validation images would not be just to provide a predictable order and to provide an easier way to map predictions to their image_id.

```
Training samples: 50916
Validation samples: 12729
Training batches: 1592
Validation batches: 398

Test samples: 29443
Test batches: 921
```

Figure 2.7.1 : Number of batches of all datasets

The images are loaded individually before processing including resizing, augmentation, and normalisation and after grouped up to form batches.

2.8 Confirm preprocessing Consistency and Leakage prevention

```
Training preprocessing:
Compose(
    <__main__.ResizeAndPad object at 0x000001F7ACF8E8B0>
    RandomHorizontalFlip(p=0.5)
    RandomRotation(degrees=[-15.0, 15.0], interpolation=nearest, expand=False, fill=0)
    ColorJitter(brightness=(0.8, 1.2), contrast=None, saturation=None, hue=None)
    ToTensor()
    Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
)

Validation preprocessing:
Compose(
    <__main__.ResizeAndPad object at 0x000001F7ACF7EE70>
    ToTensor()
    Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
)

Test preprocessing:
Compose(
    <__main__.ResizeAndPad object at 0x000001F7ACF7EE70>
    ToTensor()
    Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
)
```

Figure 2.8.1 : Confirm preprocessing between validation, training and testing

The preprocessing stage is applied throughout the test dataset, the training dataset as well as the validation dataset, though both the test dataset and the validation dataset is excluded from augmentation. Figure 2.8.1 shows what transformations have been applied to each dataset. Other than augmentation, the same resizing, padding and normalisation methods are applied to all datasets to

ensure the processed images are suitable for the resNet18 model.

Throughout the preprocessing data leakage had been minimised by removing the duplicates images before the 80/20 stratified training validation split in Figure 2.1. This makes sure that the split would not contain the same hashed images in both the validation and training dataset. The test set was kept separated from the training labelled dataset and is only used during the kaggle submission to submit the final prediction. These techniques helped minimise data leakage but it does not guarantee that there are no visually similar images even if the file hashes are checked for duplicates.

3. Model Selection and Pipeline Development

3.1 Model Selection

Convolutional Neural Network (CNN) is selected as the deep learning model because it is capable of learning image patterns. These patterns include edges, textures, and shapes in a picture of a dog and a cat. Convolutional filters contain learnable weights that are applied to local areas of an image to detect specific visual patterns. As images go through multiple convolutional layers, extracted features become more and more complex, thus allowing more complex patterns to be recognized and eventually used to classify. The use of convolutional filters allows the identification of the current pixel with the adjacent pixels, allowing CNN to learn spatial relationships such as the arrangement of shapes and edges. Additionally, CNN uses less weights due to parameter sharing, thus generally lowering the total number of weights compared to a fully connected neural network. These characteristics are what the dataset needs, as shown in Figure 1.4.1, where the images show varying quality, facial orientation, etc.; therefore, the model would need to recognize similar patterns across images as well as learn local pixel arrangement to identify all 8 classes.

A Transfer learning model is used, called ResNet-18; this is a pretrained CNN model based on weights pretrained on the ImageNet dataset to identify 1,000 categories. This allows the previously learned

weights to be used on the training dataset. The final classification layer would be changed from 1000 categories to 8 target classes.

ResNet18 where it provides a pathway (residual connection) between convolutional layers, unlike a basic CNN that processes data through convolutional layers in order. This shortcut is important in making sure that the gradient reduces the vanishing gradient issue where the gradient becomes increasingly smaller as it propagates back through the convolutional layers. This results in slower learning for the model, as the initial convolutional layers would receive minor weight changes. The added pathway makes sure that gradients are able to propagate back more effectively (He et al., 2016).

Compared to other models, ResNet18 is relatively lightweight compared to other larger pretrained architectures like VGG16. According to Tanvir et al. (2026), ResNet18 contains approximately 11.18 million parameters compared to VGG16's 134.27 million parameters. The model would be trained on Kaggle's GPU environment therefore has limited resources and usage; therefore, selecting a lower-parameter model is a better choice for this project.

3.2 Model Architecture

The architecture of ResNet18 starts by passing the original image (224x224 pixels) through the initial convolutional layer, which includes 3 input channels (RGB) and 64 different filters, each with a 7x7 kernel. Once the feature map is produced, batch normalisation is applied, which basically normalises the feature activation values. ReLU activation then introduces non-linearity that replaces negatives with zeros and keeps the positive activation values. Max pooling is used after this process to reduce the spatial size of the feature map.

After that process, this is followed by 4 layers; each layer consists of 2 residual blocks, each residual block contains a convolutional layer, batch normalisation, ReLU, another convolutional layer, and

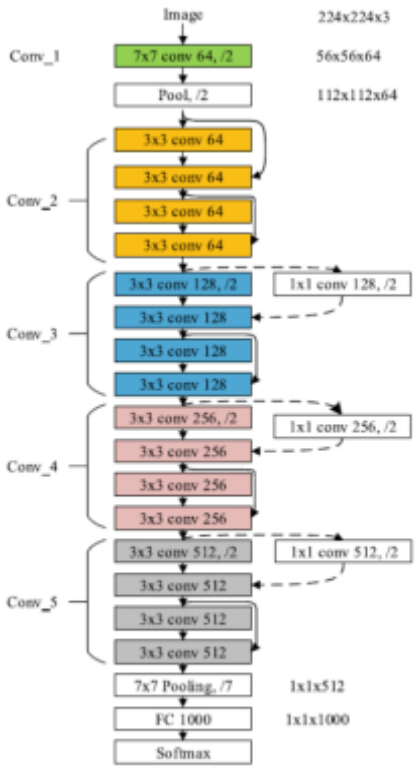


Figure 3.2.1 : ResNet18 Architecture
Al Rikabi and Muhammad (2024, Figure 2).

another batch normalisation. This residual block contains a path that skips the main convolutional layers from the input to the output feature map to add the initial input feature map. Each layer gets an increasingly higher number of feature channels (64, 128, 256 to 512) so that it can learn more complex patterns.

Once these feature maps are produced from those 4 layers, the final global average pooling is applied to the 512 feature maps to find the mean value of each. This is then flattened into a single vector and transferred to the classification layer. The fully connected classification layer converts the 512 feature values into 8 output values relating to the 8 classes in the dataset.

3.3 Transfer Learning and Model Modification

To train the model, ImageNet pre-trained weights are used for the initial weight configuration of the training process. This is because the weights are already learned from training on 1000 classes. For the initial training, the weights would remain unchanged except for the fully connected classification layer, where the weights are randomly initialized and updated during training the model. After evaluating the prediction results from the unchanged weights, the pretrained weights would be allowed to change for different layers; this verifies if optimizing the weights for this dataset improves classification compared to sticking with the pretrained weights. As discussed in tasks 3.1 and 3.2, the original classification layers are changed from 1000 outputs to 8 output layers for this dataset. This is the only modification that is made in order to preserve as much of the original ResNet18 architecture.

3.4 Model Input and Output Configuration

The table below shows the used modified ResNet18 architecture model on the input and output configuration.

Component	Configuration	Description

Input Image Size	224x224 pixels	The preprocessing would adjust the images to this size
Input Channels	3 Channels (RGB)	Red, blue, and green channels
Output Layer	1 fully connected layer with 8 outputs	Converts 512 feature maps into 8 output classes.
Activation Function	ReLU	Adds nonlinearity
Number Of Classes	8	Since there are CatEyes, DogEyes, CatFace, DogFace, CatMouth, DogMouth, CatNose, and DogNose classes.

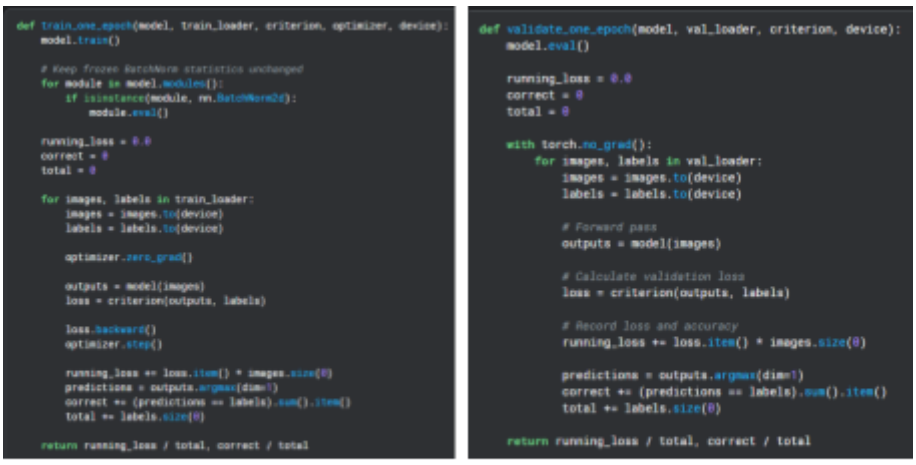
3.5 Training Hyperparameters

The table below shows the hyperparameters, methods, and values used for the initial training and validation of the model.

Hyperparameter	Selected Value	Justification
Loss Function	CrossEntropyLoss	This loss function method is used because this is a multiclass classification dataset. It allows the comparison of multiple class logits to check against the correct label class. This would produce a loss value that indicates how well the prediction aligns with the true label. This would help provide signals for updating weights.
Optimiser	Adam (Adaptive Moment Estimation)	This was selected because it utilizes previously calculated gradients and current gradients to update individual weights, since it helps decide the direction and size of what an individual weight normally receives to determine the next weight update. This improves

		efficiency and stability since it does not solely rely on the current gradient.
Learning Rate	0.001	This value is chosen because Adam and a 0.001 learning rate often pair together. If the learning rate is too high, it may cause an overly updated weight, which causes instability, while a low learning rate would be incredibly slow to train the model. This isn't necessarily an optimal choice, but it is a baseline until further training and validation loss is evaluated.
Batch Size	32	Determining the batch size is experimental, but since a Kaggle notebook would be used for training, a balance between training efficiency and memory consumption would be needed. Therefore, 32 is used as the starting point.
Number Of epochs	10	The number of epochs is set to 10 as the baseline value. This is experimental, as the validation results would need to be evaluated for overfitting or underfitting before changing the number of epochs.

3.6 Training Pipeline Implementation



```
def train_one_epoch(model, train_loader, criterion, optimizer, device):
    model.train()

    # Keep frozen BatchNorm statistics unchanged
    for module in model.modules():
        if isinstance(module, nn.BatchNorm2d):
            module.eval()

    running_loss = 0.0
    correct = 0
    total = 0

    for images, labels in train_loader:
        images = images.to(device)
        labels = labels.to(device)

        optimizer.zero_grad()

        outputs = model(images)
        loss = criterion(outputs, labels)

        loss.backward()
        optimizer.step()

        running_loss += loss.item() * images.size(0)
        predictions = outputs.argmax(dim=1)
        correct += (predictions == labels).sum().item()
        total += labels.size(0)

    return running_loss / total, correct / total
```

```
def validate_one_epoch(model, val_loader, criterion, device):
    model.eval()

    running_loss = 0.0
    correct = 0
    total = 0

    with torch.no_grad():
        for images, labels in val_loader:
            images = images.to(device)
            labels = labels.to(device)

            # Forward pass
            outputs = model(images)

            # Calculate validation loss
            loss = criterion(outputs, labels)

            # Record loss and accuracy
            running_loss += loss.item() * images.size(0)

            predictions = outputs.argmax(dim=1)
            correct += (predictions == labels).sum().item()
            total += labels.size(0)

    return running_loss / total, correct / total
```

Figure 3.6.1: Training and validate one epoch functions

The training pipeline was implemented in the Kaggle notebook using the accelerator GPU T4 x2. In the notebook, the preprocessing, training, and validation dataset were loaded from task 2 through the created

data_pipleine.py module that creates the dataloader for the training and validation. The ResNet18 architecture model was then loaded and configured with the task 3.5 specified hyperparameters.

Two functions have been created to facilitate the training, train one epoch and validate one epoch. The train one epoch function contains the process each batch of 32 images would pass through the ResNet18 model to generate the 8 classification values. These values are used to calculate loss from the implemented crossEntropyLoss. Then backpropagation was used to calculate the gradient in which the optimizer would update the trainable weights of the fully connected classification layer.

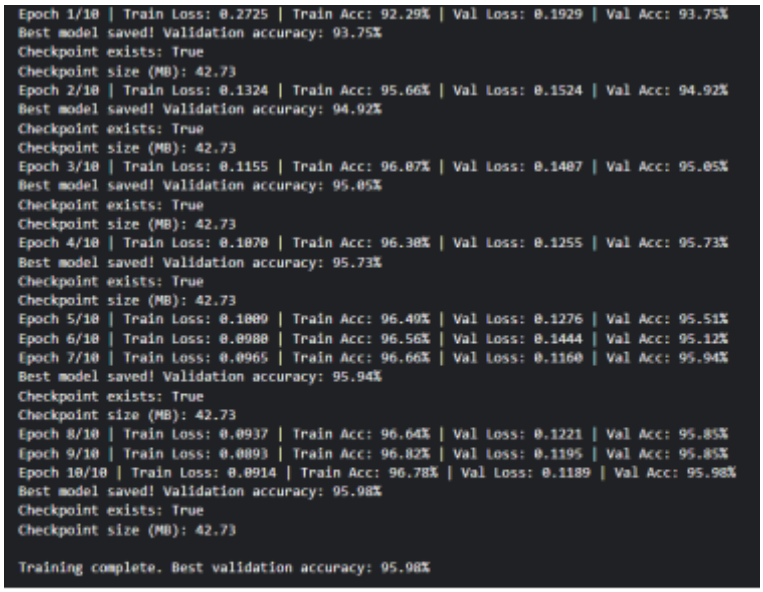


Figure 3.6.2: First model train result

The Validation process function, contains the process of validating one epoch from the trained weights of its corresponding training epoch run. The model gradient calculations were disabled to prevent weight updates.

The training and validation accuracy were recorded through the 10 epochs and the model train weights were saved whenever there is an improvement in validation accuracy so this can be reused for the Kaggle test prediction.

3.7 Kaggle Test Prediction Pipeline

To prepare a submission prediction for the test dataset, the ResNet18 model trained with the highest validation accuracy was saved and loaded to generate predictions on the unlabelled test dataset images. The model was placed into the same position as the validation step for 1 epoch, where no weights are changed because the gradient calculations are disabled. Each image is passed through the model to produce the 8 output logits, where the highest logit value is the predicted class. These values were converted into their corresponding class name to be prepared for storage in the submission.csv file.

In the submission.csv file, the predictions were then matched with their image ID and stored in the submission.csv file following the same format as the sample_submission.csv.

```
Total predictions: 29443
Total image IDs: 29443
First five predictions: ['DogEye', 'DogMouth', 'DogNose', 'DogFace', 'CatMouth']
```

Figure 3.7.1: First test dataset prediction

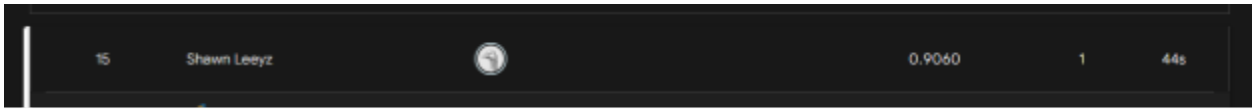


Figure 3.7.2: First submission uploaded to public Kaggle Leaderboard (16/10/2026)

4. Training Strategy and Overfitting/Underfitting Management

4.1 Training Strategy

As discussed from section 3, the training of the dataset is completed using pretrained ResNet18 CNN model, the fully connected classification layer was the only layer that was allowed to change in weights

while the weights in the convolutional layers were frozen in order to test whether pretrained weights learned from the ImageNet has suitable weights that do not require updating.

The process was performed over 10 epochs, using the Adam optimizer, a learning rate of 0.001, and a batch size of 32. In each training batch of 32, training was done through forward propagation, and the cross-entropy loss measures the prediction probability between the predicted class label (8 classes) and the actual class label. Then, to update the weights, backpropagation was performed to compute the gradients from the model's learning during training. The optimizer uses the gradient to update the fully connected classification layer.

After every epoch run, the model was also validated against the validation dataset. The model included the training and validation accuracy as well as the loss value of both the training and validation sets. The weights are saved when the validation accuracy improves into a pth file for Kaggle submission after all 10 epochs are run.

4.2 Training and Validation Performance

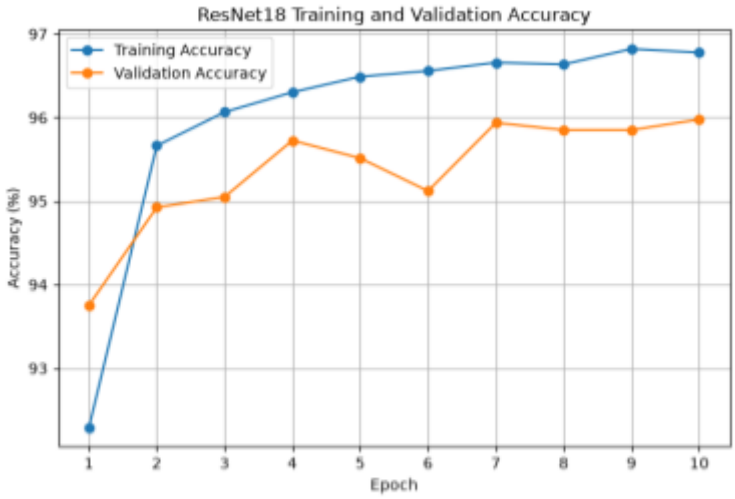


Figure 4.2.1: Learning curve of initial training for training and validation accuracy

Figure 4.2.1 shows the results of the training and validation accuracy across the 10 epochs from the initial training of the pretrained ResNet18 model on the training dataset. The figure shows that the largest improvement for both the training and validation accuracy occurred during the first and second epochs, where the accuracy increased from 92.29% for training and 93.75% for validation to 95.66% for

training and 94.92% for validation. The increases were lower for the epochs from 3 to 9 for validation accuracy. Around the 10th epoch is when the validation and training accuracy stabilizes. Unlike the gradual increase in training accuracy across the 10 epochs, the validation accuracy experienced several fluctuations. This is seen with the 4th epoch, when it reaches 95.73% before decreasing to 95.51% in the 5th epoch and 95.12% in the 6th epoch, then increasing from 95.12% to 95.94% in the 7th epoch. Therefore, this means that the validation accuracy improvement does not always result in an improvement in validation accuracy, though the overall validation accuracy did improve.

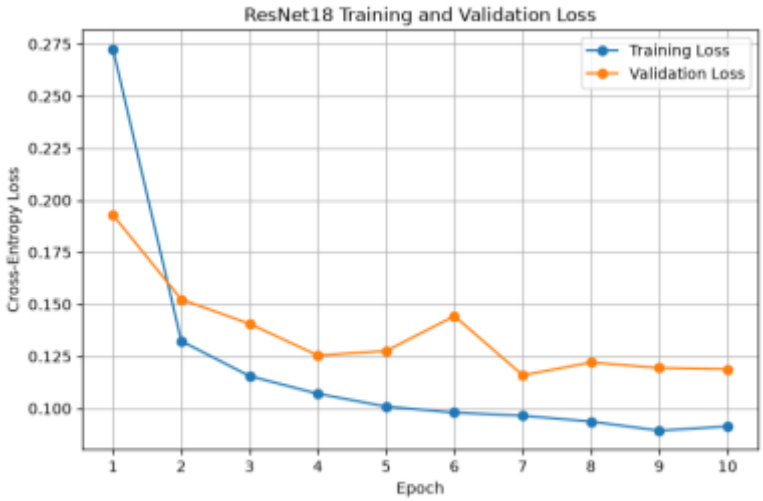


Figure 4.2.2: Learning curve of initial training for Training and validation loss

Figure 4.2.2 shows the training and validation loss with the same x-axis (epoch) as the training and validation accuracy curve. At the start, the training loss decreased from 0.273 in the first epoch to 0.0914 in the 10th epoch. The largest change in loss for training was at the start, between epochs 1 and 2, and gradually decreased and approached convergence as the learning continued for the fully connected classification layer. For the validation loss, it experiences the same fluctuation as the validation accuracy graph, but instead it trends down. This is demonstrated between epochs 4 and 6, as it increased from 0.13 to 0.14 before decreasing to 0.116 at epoch 7. The overall validation loss decreased from 0.1929 to 0.1189, which still indicates that the model predictions generally reduce loss for the validation dataset.

4.3 Overfitting and Underfitting analysis

From the analysis of the Training and validation accuracy and loss learning curve in Figures 4.2.1 and 4.2.2, the ResNet18 model used to train did not show significant overfitting and underfitting through the 10 epochs.

This is concluded because the final training accuracy was 96.78% and the final validation accuracy was 95.98%. The difference is 0.8 percent, which is relatively small. Therefore, this does not provide strong evidence that the initial training was overfitting.
However, the epochs between 4 and 6 increased in validation loss, which would mean that during those learning periods the model wasn't learning patterns, which could indicate some temporary overfitting, but the validation loss decreased from there. This indicates that the increase in loss wasn't a constant issue.

The training was not clearly shown to be underfitting because the model achieved a high training accuracy, and the low training loss remained low through the epochs, which supports this argument. Therefore, this showed that the model was able to genuinely learn patterns in the training dataset and applied the learned patterns to the validation dataset without a significant accuracy difference. Overall, the results suggested that the training model was able to generalise reasonably well to the validation dataset, though this could be different for unseen data.

4.4 Techniques to Improve Generalisation

In the initial training of the model, there are 2 generalisation techniques applied, this being data augmentation and transfer learning. For data augmentation, changes have been applied to the training dataset images to improve generalization. This is done by applying a 50% probability of random flipping, rotating the image by plus or minus 15 degrees, and changing the brightness by plus or minus 20%. This is applied through the preprocessing stage as discussed in section 2.6. Implementing these

changes to the image allows the model to not rely so much on the exact appearance of the images, which would help with unseen images.

Transfer Learning was also implemented by ResNet18 pretrained on ImageNet that has 1000 output classes. For the initial training and prediction of the model, the convolution layers' weights were not changed; only the fully connected classification layers were changed. Therefore, this reduces the number of trainable weights and allows the model to reuse learned weights from ImageNet.

The fine tuned model changes, the weights gotten from the initial training model was reused and added an additional 5 epochs of training instead of retraining from the original fully connected classification layer weights from ImageNet. A low number of epochs was chosen to further refine the existing weights while reducing the needed training time and reducing the risk of overfitting. Additionally, the final layer weights (the last group of convolution layers), where the model recognizes more complex patterns, were allowed to change. Therefore, this made the learning suitable for this dataset instead of utilizing the ImageNet-trained weights. Furthermore, the learning rates were reduced to 0.00001 for the 4th layer and 0.0001 for the fully connected classification layer to prevent large changes to happen to the existing weights.

4.5 Comparison and Justification



Figure 4.5.1: Learning curve of fine-tuned training for training and validation accuracy

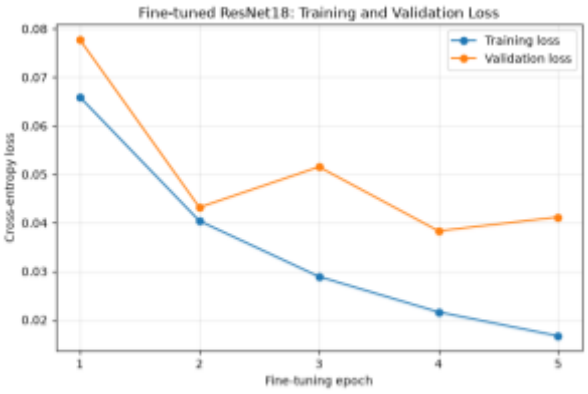


Figure 4.5.2: Learning curve of fine-tuned training for Training and validation loss

The results shown after applying fine-tuning to the original baseline training model for an additional 5 epochs showed that the highest validation accuracy increased from 95.98% in the initial training to 98.76% for the changed model; this resulted in an improvement of 2.78%. This means that the applied changes had a positive effect and that the last group of convolution layer updated weights have helped in the model's adaptability to learn features on the validation dataset.

With the increase in validation accuracy, the corresponding training accuracy increased as well from 97.74% at the first epoch to 99.41% at epoch 5. The validation accuracy did experience fluctuations similar to the initial model; the fine-tuned model seemed to have greater fluctuation but that is because it is trained across 5 epochs not 10 like the initial model which would make the fluctuation less highly noticeable. The highest value reaches 98.76% at epoch 4 before decreasing to 98.63% at epoch 5. This occurred similarly with the validation loss increase from 0.0383 to 0.0412 between epoch 4 and 5.

This could indicate that overfitting might have occurred as training performance improved while validation performance slightly decreased but this only occurred through a 1 epoch difference of only 1 epoch. To further investigate whether overfitting was occurring, the fine-tuning process could be extended beyond five epochs to observe whether training accuracy continued increasing while validation accuracy decreased and validation loss increased.

5. Model Evaluation and Kaggle Submission

5.1 Validation Set Evaluation

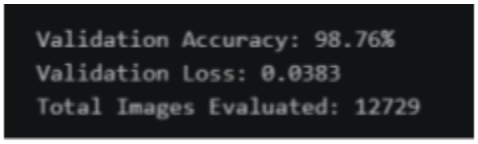


Figure 5.1.1: Validation accuracy and loss output

The final and highest validation accuracy achieved by the fine-tuned ResNet18 model across the 12,729 validation dataset images was 98.76%, with a cross-entropy loss of 0.0383 (shown in Figure 5.1.1).

This suggests that the predictive capabilities of the model are strong across the 8 classes, and the cross-entropy loss means that there is little difference between the true labels and the predicted labels.

	precision	recall	f1-score	support
CatEye	0.9874	0.9903	0.9888	2775
CatFace	0.9978	0.9986	0.9982	1389
CatMouth	0.9957	0.9950	0.9953	1388
CatNose	0.9883	0.9733	0.9808	1388
DogEye	0.9675	0.9888	0.9780	2315
DogFace	0.9974	0.9896	0.9935	1158
DogMouth	0.9940	0.9957	0.9948	1158
DogNose	0.9902	0.9637	0.9768	1158
accuracy			0.9876	12729
macro avg	0.9898	0.9869	0.9883	12729
weighted avg	0.9877	0.9876	0.9876	12729

Figure 5.1.2: precision, recall and F1-score across 8 classes

Additional metrics have been evaluated alongside validation accuracy and validation loss, these include precision, recall, f1-score. The overall model achieved a macro-average precision of 98.98%, recall of 98.69%, and f1-score of 98.83%. These results show that across the 8 classes the overall precision and recall were high. However the most notable result from the 8 classes is the DogEye class with precision of 96.75% therefore indicates that the prediction ability for Dog eye isn't as capable as the other classes. This would be further analysis in section 6. The f1 score which shows the balance between precision and recall and reducing the effect of class imbalance, where CatFace scored the highest f1-score of 99.82% while DogNose had the lowest f1-score at 97.68%.

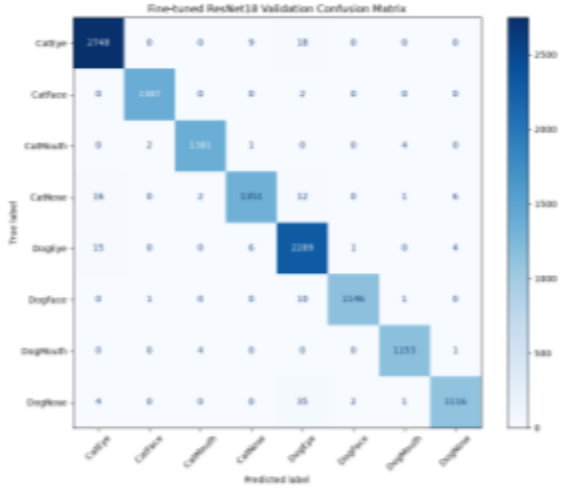


Figure 5.1.3: Confusion Matrix of all 8 classes

The confusion matrix further helped provide much more detail in comparison to just percentages in Figure 5.1.2. Figure 5.1.3 shows the individual predictions made across all 8 classes. The confusion matrix clearly showed that the majority of predictions match the true label which relates to the overall high macro average from Figure 5.1.2. Exploring the DogEye which had the lowest precision score, the confusion matrix showed that even though the majority of the predictions were predicted correctly, there are 35 predictions as DogEye which is the true label is supposed to be DogNose. This is the highest false prediction made compared to under false prediction class. This would be further discussed in task 6 to find out why DogEye in particular receives the highest false

prediction compared to the other classes. Another noticeable observation is that CatEye received incorrect predictions from other classes, including 16 CatNose images and 15 DogEye images being classified as CatEye. These false-positive predictions contributed to the reduction in CatEye's precision.

5.2 Kaggle Test Prediction and Submission

To generate the prediction to submit into the Kaggle competition, the process involves loading the test dataset, generating predictions, creating the submission file and submitting.

submission_finetuned.csv > data

```
1   image_id,class
2   1,DogEye
3   2,DogMouth
4   3,DogNose
5   4,DogFace
6   5,CatMouth
7   6,DogFace
8   7,CatMouth
9   8,DogEye
10  9,CatEye
```

Figure 5.2.1: Fine-tuned model submission format

The unlabelled test dataset was loaded with the same preprocessing as the validation dataset. To predict, the saved model (with the trained weights) was loaded using the saved checkpoint from the training. Then the predictions were created by selecting the highest-scoring class and converting the numerical class indices into their corresponding class name. After this, the prediction was matched with the image ID to create the same format as the "sample_submission.csv". The original RestNet18 prediction was submitted through the "submission.csv" file and the fine-tuned model was submitted through the "submission_finetuned.csv", both are then uploaded and submitted to the Kaggle competition.

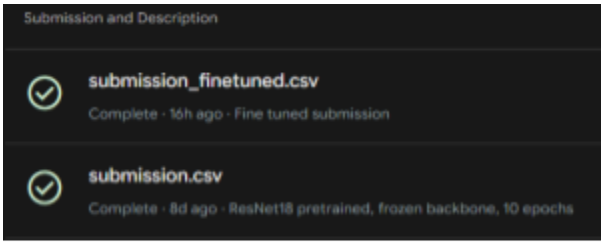


Figure 5.2.2: Confirmation of Submission to Kaggle

5.3 Kaggle Leaderboard Results



Figure 5.3.1: Kaggle Submission Accuracy Result (Initial)



Figure 5.3.2: Kaggle Submission Accuracy Result (fine-tuned)

After submitting the predictions of both the initial model and the fine-tuned model to predict the public hidden dataset, figure 5.3.1 and figure 5.3.2 showed the results. The initial model achieved an accuracy of 90.60% while the fine-tuned model received an accuracy score of 94.14%. This results in an improvement of 3.54%. Therefore the modification from the initial model showed an improvement in classification performance on the public leaderboard dataset although the result may vary due to the randomness of training.

5.4 Validation and Kaggle Performance Comparison

Comparing the validation results with the Kaggle results, both the initial model and fine-tuned model achieved a lower accuracy on the Kaggle dataset than on the validation dataset. For the initial model, validation accuracy was 95.98%, while the accuracy for the Kaggle dataset was 90.60% (a difference of 5.38%). The fine-tuned model had a validation accuracy of 98.76%, while the Kaggle dataset had 94.14% (a difference of 4.62%).

The first possible reason that could result in this difference, is that it's caused by the variation between the datasets including variation of characteristics harder to recognize dogs and cat facial features. Additionally, a possible reason is that the checkpoint was saved only when the validation accuracy result reached its highest during training; therefore, it could mean that it was able to make better

prediction decisions on the validation dataset than on an unseen dataset like the public Kaggle dataset. These reasons cannot be confirmed but are just educated speculations.

However, the validation accuracy was higher than the Kaggle dataset accuracy. The improvement seen from the initial to the improved model was still valuable, as it showed that it was able to adapt its features more closely to the 8 different classes, resulting in better performance on the unseen dataset.

6. Results Analysis and Improvement Strategies

6.1 Overall Performance Analysis

The initial model has already achieved a high validation accuracy of 95.98%, even though only the weights of the fully connected classification layers were allowed to update after each batch. To suggest that the pretrained weights from the training from ImageNet already is effectively extracting the features of the dog and cat facial feature classification dataset. Though Figure 4.2.1 showed that the validation accuracy had begun to fluctuate during the later epochs, this could be a sign that it's slowly reaching convergence with the initial model configuration; this would mean that continuing training from this configuration would not yield significant results and that the configuration of the model would need to be altered in order to reach higher accuracy than currently. However, further training and introducing a higher number of epochs would be needed to confirm this.

After unfreezing the 4th group of convolution layers, from Figure 5.1.1, the validation accuracy increased from 95.98% to 97.41% for the first epoch, with the highest accuracy reaching 98.76% on epoch 4. More importantly, the cross-entropy loss decreased massively from 0.1189 to 0.0383 when compared to the initial model. This indicates that the accuracy improvement is not only limited to correctly classifying the images but also means that it is generally assigned higher probabilities to the correct classes. This

could explain the change by allowing higher-level features to be extracted and adapt more closely to the 8 classes instead of having fixed pre-trained weights from the initial model.

The Kaggle results showed that the improved model did increase the unseen Kaggle dataset accuracy from 90.60% to 94.14%, which is evidence that the improvement not only affected the validation dataset but also the unseen dataset. However, this does not mean that allowing the 4th group convolution to change is the only cause of the increase in accuracy, as multiple other configurations were changed as well. This includes the additional 5 training epochs used on the initial model's trained weights, and the lower learning rates. A separate experiment would be needed to test and isolate the unfreezing of the 4th group layer change to determine whether this change alone had the biggest impact.

6.2 Misclassification

As shown in Figure 5.1.3 in the confusion matrix, one of the most frequent misclassifications was between DogNose and DogEye. The confusion matrix showed that 35 images were predicted as DogEye, but the true label was DogNose. From this result, it showed that the errors were not distributed evenly across the 8 classes but contained a higher concentration of incorrect DogNose predictions.

Another observation made was that the misclassification involves multiple types. For example, there were 35 images classified as DogEye but were actually DogNose and 16 classified as CatEye but were actually CatNose. Both of these indicate that the model was able to identify the type of animal but wasn't able to identify the correct facial feature. The other type involves misidentifying animal types; for example, 15 DogEye images were misclassified as CatEye. So, from this evaluation, the majority of misidentified labels can't be treated as just misidentifying 1 type, as it involves misidentifying both the animal type and facial feature type.

Supporting this analysis of misclassification, Figure 5.1.2 showed that DogNose had the lowest recall of 96.37%, while DogEye had the lowest precision of 96.75%. This result relates to the confusion matrix, as the 35 DogNose images contributed to the lower recall of DogNose as well as the lower precision of

DogEye because the model incorrectly predicted DogNose images as DogEye. Furthermore, DogNose and DogFace both had 1,158 true labelled images in the validation dataset, but DogFace has a higher recall of 98.96%. This means that the classification performance can't be explained by the number of images in both classes.

A possible explanation of these errors occurring is that there are overlapping facial characteristics in an image as shown in Figure 1.4.1. This could influence the model's prediction. This reason is unverified, as the confusion matrix only shows which classifications were incorrect; it does not identify what pattern influences a particular prediction. Therefore, the images that were falsely predicted need to be examined to determine whether those images contain multiple facial features that support this explanation.

6.3 Model Limitations

There are two possible limitations identified: the first being that the remaining 3 groups of convolution layers were kept frozen with the pretrained ImageNet weights, unlike the last group of convolution layers and the fully connected classification layer. Although the result showed that keeping the pretrained weights unchanged proved to be effective, it could hinder the model's ability to adapt the lower-level and intermediate feature extraction to the 8 output classes. Though this does not mean allowing all layers to update its weights as this could cause the risk of overfitting.

As discussed in section 6.2, the overlapping facial feature was a possible performance limitation. In the preprocessing stage, the images were resized and padded, and we tried to preserve as much of the original image content as possible. Therefore, surrounding details were retrained, which could influence the model's prediction even though those details were not the intended facial feature to be classified. Therefore, the model could have learned to rely on the image surroundings as well as the center image to make a prediction. However, this wasn't confirmed by how the misclassified images were predicted by the model.

6.4 Evaluation of Implemented Decisions

In the development of the preprocessing phase, the resizing method of implementing padding towards a set size of 224x224 was chosen to prevent the image from being stretched. This allowed for the image to be preserved, this is because during the explorative stages there are some images that had important features such as a dog eye being close to the border of the image. This method decision was applied because before trying to crop or remove any original image content it was better to retain it and test on whether maintaining the details would not cause any issues in prediction. The imageNet normalization was also applied during preprocessing in order to match the normalization expected when using pretrained ResNet18 weights. Though the model did achieve a high validation accuracy, the contribution of the preprocessing stage cannot be decided since no other alternative preprocessing methods were tested.

During data augmentation, the training dataset applied brightness level changes, horizontal flipping and rotating the image. This was introduced to make the training dataset more generalized as from the exploratory stage, some images were identified to have qualities that were somewhat difficult to see the facial features, therefore this was applied to help with that. Though there was no test applied with or without the applied data augmentation, thus could not confirm its effectiveness on generalisation.

Implementing pretrained ResNet18 was as shown by the results a successful choice as the initial model had achieved a 95.98% validation accuracy while only updating the fully connected classification layer for the initial model. This implies that the usage of the pretrained extracted features from ImageNet was already useful for this 8 output classes task. From section 3.1, this model was chosen because of its smaller architecture compared to other RestNet variants, thus a lower computational requirement which can result in a lower training time. Since no other architecture was used to compare with the ResNet18 model, there isn't a comparison that could be made to check whether a similar parameter model would be more effective in predicting than the currently used model.

The training strategy, the initial experiment of freezing all weights except the fully connected classification layer, proved that this method still achieved a high validation accuracy. Therefore, updating the entire feature extractor was not necessary to reach a high validation accuracy. However, the later epoch stages showed that the validation accuracy was least likely to increase and might be limited to the same model configuration as the number of epochs increases. The fine-tuned model increased the number of trainable weights and computational requirements, but the addition of updatable weights for the 4th group of convolution layers was associated with increasing the previous model results. This result was achieved with a combination of lower learning rates; this was implemented because, since the pre-trained weights were already effective, as well as the classification layer, the decision was to make the weight updates more minor and more controlled, thus preserving the pre-trained information while allowing the weights to adapt. Although this method was successful, there is currently no comparison that could be made to determine whether reducing learning rates is most effective and whether unfreezing group layer 4 is effective; these would need to be experimented with in isolation from all the individual configuration changes in order to accurately identify which change has the biggest impact.

6.5 Proposed Improvements

There are 3 experimental improvements that could be performed and determine whether the existing refined models are effective and what possible improvement could be made. These proposed improvements include:

- **Unfreezing the 3th convolution group layers**

This is the next step of unfreezing the 4th layer in the refined model as discussed in section 6.3, the earlier convolution layers's weights were locked from being updatable, which could restrict how much the feature extraction could adapt the the 8 classes and as seen from allowing weights to change for the 4th layer this could be a plausible improvement that could be made based on the current evidence of results. Conversely this could increase the possibility of overfitting and would increase the computational requirement. This would need to be evaluated with the fine-tuned model with the same

model configuration (same epoch, same learning rate etc) to determine whether it improves solely on allowing the weights to update for the 3th convolution group layer.

- **Different resizing implementation**

As identified in section 6.2, 35 of the 42 incorrectly classified DogNose Images were predicted as DogEye. Therefore this suggests that examining these images could help determine whether implementing a different type of resizing would be appropriate to remove surrounding facial features in the image. If patterns were found during investigation that these images contain other conflicting facial features, applying a more targeting cropping method around the intended facial feature could be utilized instead of the current resizing implementation. This would need to be applied appropriately though because cropping could remove important information thus removing the models ability to extract useful patterns.

- **Changing learning rates**

The current model uses a learning rate of 0.00001 for the 4th layer and 0.0001 for the classification layer. This implementation showed that it was also part of the reason why the fine-tuned model was achieving a higher validation accuracy and accuracy on the Kaggle unseen dataset. Although this cannot be individually contributed to just the learning rate as from the initial model there were multiple changes. Therefore, experimenting with the learning rate, such as changing to a lower learning rate of 0.000001 or increasing the rate to 0.0001, could be tested, and other training sets are unchanged.

6.6 Reflection of Solution

The solution produced by the refined model clearly showed that there are strengths in this training model implementation, particularly that the final ResNet18 model was able to reach high classification accuracy while retaining most of the parameters. The use of transfer learning provided a good starting point rather than training from a completely custom CNN model. A lightweight model like ResNet18 could still achieve relatively strong results while balancing the number of parameters used in the model and performance. Additionally, the model achieved a high recall and precision score for all classes,

which meant that the model did appropriately extract important patterns that were able to identify the majority of the facial features between cats and dogs.

An issue found was that when the model makes a prediction error, it often clusters towards a class out of 8 classes. This could mean that the prediction is being influenced by surrounding facial features preserved during preprocessing. Furthermore, the difference in validation performance on the Kaggle public leaderboard was not similar, as the Kaggle test subset accuracy was lower than the validation result. This raises uncertainty about the consistency of the model to perform on an unseen dataset because both the initial model and fine-tuned model performed worse on the Kaggle unseen dataset than on the validation dataset.

Through this reflection, instead of multiplying multiple configuration changes in a single training model integration, for better analysis and understanding, one configuration at a time should be tested and the results evaluated. This would provide stronger and clearer evidence of which had a positive impact and identify which is a more suitable configuration, unlike the current method of evaluation, as most of the results had to be based on educated guesses. Therefore when conducting future experiments using deep learning models and in this other kaggle competition, The current approach on experimenting would be swapped for the stated approach above when developing the model further.

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Github

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